
2019-Ph-H1-Q21

Section: Particles and Waves

Topic: The Inverse Square Law

Question:

If the distance from a point source of light is doubled, what happens to the intensity?

Answer:

By the inverse square law:

$$I \propto \frac{1}{r^2}$$

So, if distance r doubles, intensity becomes:

$$\frac{1}{(2)^2} = \frac{1}{4}$$

Correct Option: D (It becomes a quarter)

Guidance for Students:

Inverse square law applies to light, sound, gravity, radiation intensity, etc.

Revision Tips:

- Double the distance $\rightarrow$ intensity $\div 4$
 - Halve the distance $\rightarrow$ intensity $\times 4$
 - Know and use $I \propto \frac{1}{r^2}$
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